ISOMERISM

- **What is isomerism:-**

The compounds which have the some molecular formula but differ from each other in physical or chemical properties are called isomer and the phenomenon is called isomerism.

There are two main type of isomerism

- Structural isomerism
- Stereo isomerism

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- **1. Structural Isomerism:-** When the isomerism is due to difference in the arrangement of atoms within the molecules without any reference to space is called structural isomerism.
- **(a) Chain isomerism:** Structural isomerism is of five type chain isomer have the same molecular formula but differ in the order in which the carbon atoms are bonded to each other.
- **Example :** n-Butane and isobutane

$$CH_3—CH_2—CH_2—CH_3$$

n-Butane

$$\begin{array}{c} CH_3 \\ | \\ CH_3—CH—CH_3 \end{array}$$

2 Bromobutane

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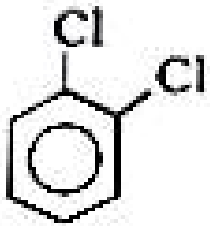
- **(b) Position isomerism:-** In this isomer have the same molecular formula but differ in the position of a functional group on the carbon chain.
- **Example :** 1-Bromobutene and 2- Bromobutane

$CH_3—CH_2—CH_2—CH_2Br$

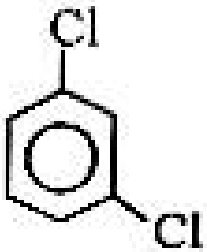
1-Bromobutane

$CH_3—CH(Br)—CH_2—CH_3$

2 Bromobutane



o-dichlorobenzene



m-dichlorobenze

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- **Functional isomerism:-** Functional isomer have the same molecular formula but different functional groups.

$CH_3—CH_2—OH$

Ethyl alcohol

$CH_3—O—CH_3$

Dimethyl ether

$$CH_3—\overset{\overset{\displaystyle O}{\|}}{C}—CH_3$$

Acetone

$$CH_3CH_2—\overset{\overset{\displaystyle O}{\|}}{C}—H$$

Propionaldehyde

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- **(d) Metamerism :** This type of isomerism is due to the unequal distribution of carbon atoms on either side of the functional group.

  Example: Diethyl ether and methyl propyl ether

  $CH_3CH_2–O–CH_2CH_3$ $CH_3–O–CH_2CH_2CH_3$

  Diethyl ether Methyl propyl ether

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Stereoisomerism

- When isomerism is caused by the different arrangement of atoms or groups in space the phenomenon us called stereoisomerism.
- The stereo isomers have the same structural formulas but differ in arrangement of atoms in space.
- Stereo isomerism is of two types
- Geometrical or cis-trans isomerism
- Optical isomerism
- **(a) Geometrical isomerism :-** This type of isomerism is due to restriction in rotation about double bonds or about single bonds in cyclic compounds.

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Geometrical isomerism in Alkenes

- The carbon atoms of the carbon-carbon double bond are Sp^2 hybridized.
- The carbon-carbon double bond consists of a σ bond and a π bond.
- The σ bond is formed by the overlap of Sp^2 hybrid orbital. The π bond locks the molecule in one position.
- The two carbons of the C = C bond and the four atoms that are attached to them lie in one plane and their position in space are fixed.
- Therefore rotation around the C = C bond is not possible because rotation would break the π bond.

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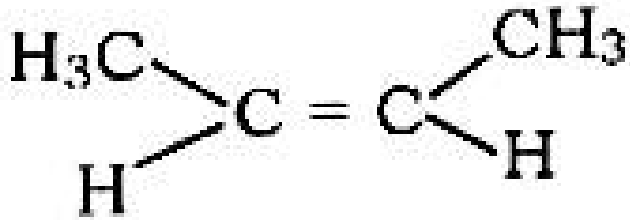
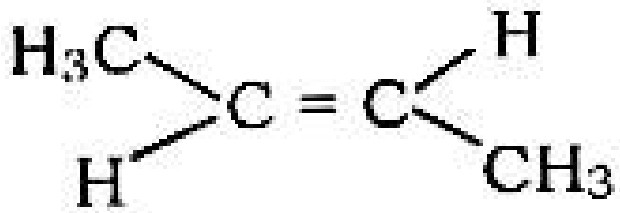
- This restriction of rotation about the carbon-carbon double bond is responsible for the geometrical isomerism in alkenes.

Example : 2-Butene. It exists in two special arrangement



Cis-2-Butene



trans-2-Butene

- The cis isomer is one in which two similar groups are on the same side of the double bond.
- The trans isomer is that in which two similar groups are on the opposite sides of the double bond.

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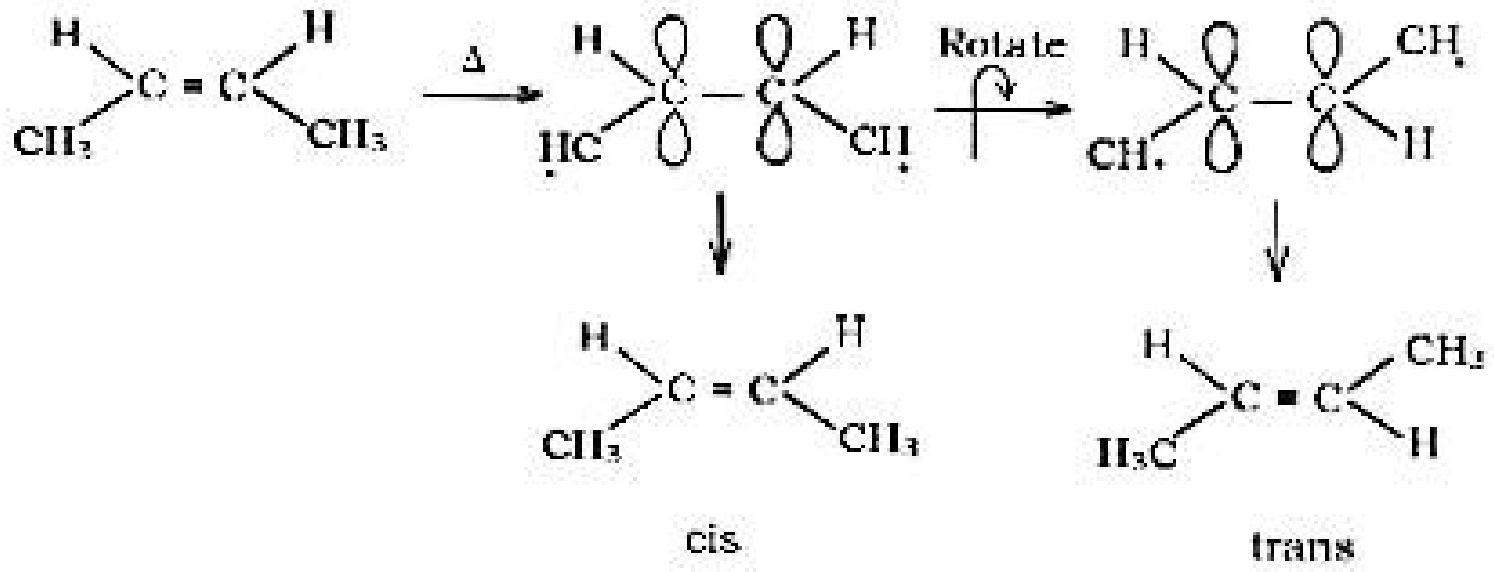
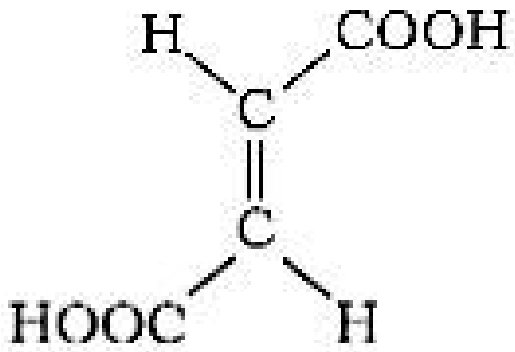
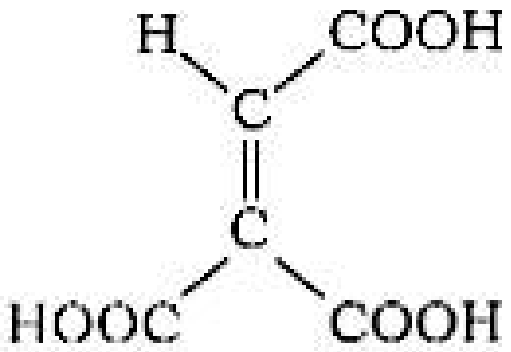
- The conversion of cis isomer is heated to a trans isomer is possible only if either isomer is heated to a high temperature or absorb light.
- The heat supplies the energy to break the π bond so that rotation about σ bond becomes possible on cooling, the reformation of π bond can take place in two ways giving mixture of trans-2-butene and cis-2-butene.



- **Example:** Butenedioic acid (HOOC–CH=CH–COOH)



trans-Butenedioic acid
(Fumaric acid)
mp = 130°C



cis-Butenedioic acid
(Maleic acid)
mp = 288°C

- These isomers differ in physical properties such as
- melting point, dipole moment and acidities.
- The trans isomer has no dipole, it is symmetrical and cis isomer is stronger than trans isomer.

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